

Experiment 1: Critical Path Method (CPM)

Aim:

To create a project schedule using the Critical Path Method (CPM) in ProjectLibre by defining project tasks, durations, dependencies, and identifying the critical path.

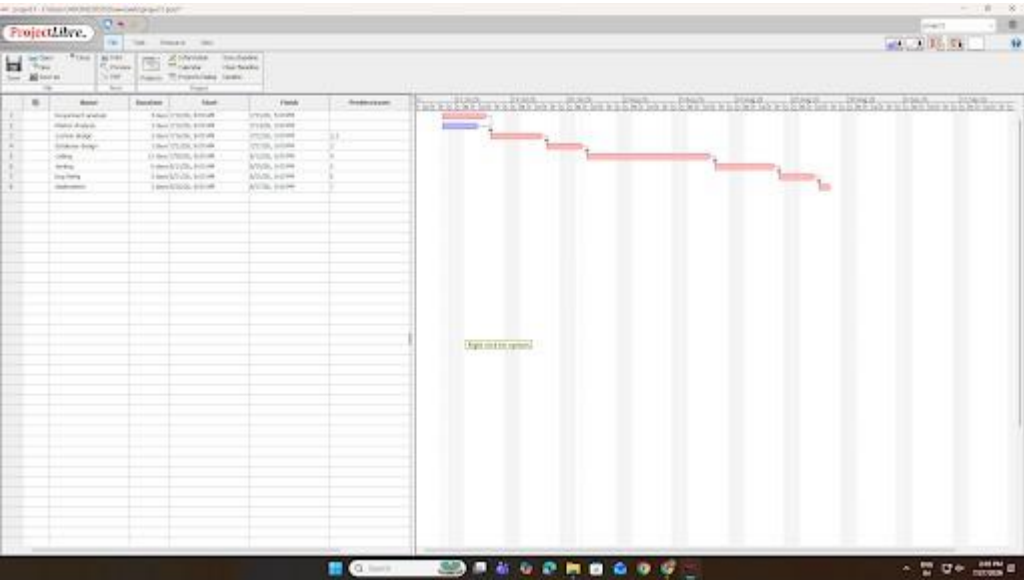
Algorithm:

1. Start ProjectLibre.
2. Create a new project by entering the project name and start date.
3. Add all project activities with their respective durations.
4. Define the dependencies between tasks.
5. Allow ProjectLibre to calculate the project schedule.
6. Identify the critical path highlighted in the Gantt Chart.
7. Save the project and observe the schedule.

Procedure

1. Open **ProjectLibre**.
2. Select **File → New** and create a new project.
3. Enter the project details such as project name and start date.
4. In the Gantt Chart, enter the list of activities and their durations.
5. Specify the predecessor (dependency) for each activity.
6. ProjectLibre automatically calculates the start date, finish date, slack, and critical path.
7. View the Gantt Chart and Network Diagram to identify the critical path.
8. Save the project file.
9. Capture the output (Gantt Chart/Network Diagram) for documentation.

OUTPUT:



The screenshot displays the ProjectLibre software interface with the Resource sheet view. The table lists resources with columns for Name, RBS, Type, E-mail Address, Material Label, Units, Group, Max. Units, Standard Rate, Overtime Rate, Cost Per Unit, and Account At.

ID	Name	RBS	Type	E-mail Address	Material Label	Units	Group	Max. Units	Standard Rate	Overtime Rate	Cost Per Unit	Account At
1	john		Work		0			100%	\$0.00/hour	\$0.00/hour	\$0.00/hour	15
2	leah		Work		1			100%	\$0.00/hour	\$0.00/hour	\$0.00/hour	15
3	mary		Work		2			100%	\$0.00/hour	\$0.00/hour	\$0.00/hour	15
4	tom		Work		3			100%	\$0.00/hour	\$0.00/hour	\$0.00/hour	15
5	carol		Work		4			100%	\$0.00/hour	\$0.00/hour	\$0.00/hour	15